
Project Weekly Report 3

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1 Weekly Progress Overview

Diffusion Q-Learning (Diffusion QL) presents several challenges:

- High computational costs due to the need for full Markov chain propagation during training.
- Incompatibility with likelihood-based RL algorithms stemming from the intractable likelihoods of diffusion models.
- Difficulty in online fine-tuning with state-of-the-art algorithms like PPO, primarily because the required probability ratio $r_t(\theta)$ for policy updates is not readily computable.

While our previous work on EDP (detailed in Report 2) addressed the first two limitations, this week's focus was on resolving the third challenge through the development of DPPO.

2 Brief Introduction to DPPO

Diffusion Policy Policy Optimization (DPPO) is an algorithmic framework for fine-tuning pre-trained diffusion-based policies in continuous control and robot learning using policy gradient methods from reinforcement learning. While PG methods were previously conjectured to be inefficient for diffusion policies in control tasks, DPPO demonstrates strong performance and efficiency.

The core idea of DPPO is to treat the denoising diffusion process as an internal Markov Decision Process (MDP), termed the "Diffusion MDP". This Diffusion MDP is then embedded within the standard environment MDP, forming a two-layer "Diffusion Policy MDP" ($\mathcal{M}_{DP}$). The policy within this $\mathcal{M}_{DP}$ has an analytically tractable Gaussian likelihood at each denoising step, making it amenable to PG updates. DPPO specifically instantiates this with Proximal Policy Optimization (PPO).

Key aspects of DPPO include:

- An advantage estimator that respects the two-level MDP structure, incorporating an environment discount γ_{ENV} and a denoising discount $\gamma_{DENOISE}$ [cite: 895, 899]. The denoising discount downweights the contribution of noisier, earlier denoising steps.
- A state-value function $V(s_t)$ that depends only on the environment state s_t , not the full Diffusion Policy MDP state, which enhances training efficiency and stability.
- Best practices such as fine-tuning only the last few denoising steps or fine-tuning DDIM sampling, and modifications to the diffusion noise schedule for stable training and exploration.

DPPO has shown benefits like structured exploration near the pre-training data manifold, training stability, and robust policy performance, even enabling zero-shot sim-to-real transfer.

3 Our Innovations

Building upon the ability of EDP and DPPO to address distinct limitations of Diffusion QL, our primary innovation is a synergistic two-stage framework that integrates these algorithms. This approach is structured as follows:

- **Offline Pre-training with EDP:** In the initial phase, EDP is employed for robust policy and value (or Q-value) function pre-training using the available offline dataset. For the purpose of this combined framework, EDP is utilized purely for its offline learning capabilities, without a subsequent online improvement phase by EDP itself.
- **Online Fine-tuning with DPPO:** Following the EDP pre-training, DPPO is deployed for online policy refinement through interaction with the environment. Critically, DPPO initializes its online phase by inheriting the policy network from EDP. Furthermore, DPPO’s state-value network ($V(s)$) is initialized based on the value (or Q-value) network learned by EDP. This provides DPPO with a more informed starting point than typical behavior cloning, which is often used for pre-training in standalone DPPO setups.

This methodology aims to capitalize on EDP’s effectiveness in leveraging offline data to establish a strong initial policy and value estimation, which DPPO then further enhances and adapts through efficient online interaction and policy gradient updates.